EXPERIMENT – 1

1. Write a Java program that implements Quick sort algorithm for sorting a list of names in ascending order

```
class Main {

   static int partition(String[] names, int low, int high) {
      String pivot = names[high];
      int i = low - 1;

      for (int j = low; j <= high - 1; j++) {
         if (names[j].compareTo(pivot) < 0) {
            i++;
            swap(names, i, j);
         }
      }

      swap(names, i + 1, high);
      return i + 1;
   }

   static void swap(String[] names, int i, int j) {
      String temp = names[i];
      names[i] = names[j];
      names[j] = temp;
   }

   static void quickSort(String[] names, int low, int high) {
      if (low < high) {
         int pivot = partition(names, low, high);
         quickSort(names, low, pivot - 1);
         quickSort(names, pivot + 1, high);
      }
   }

   public static void main(String[] args) {
      String[] names = {
         "Anvesh", "Kaushik", "Mahira", "Shweta", "Prachi", "Shubham", "Gurvir", "Ibrahim"
      };

      System.out.println("Before Sorting: ");
      for (int i = 0; i < names.length; i++) {
         System.out.println(names[i]);
      }

      quickSort(names, 0, names.length - 1);

      System.out.println();
      System.out.println("After Sorting: ");
```

```
        for (int i = 0; i < names.length; i++) {
            System.out.println(names[i]);
        }
    }
}
```

OUTPUT:

```
Before Sorting:
Anvesh
Kaushik
Mahira
Shweta
Prachi
Shubham
Gurvir
Ibrahim

After Sorting:
Anvesh
Gurvir
Ibrahim
Kaushik
Mahira
Prachi
Shubham
Shweta
```

Figure 1: Output of program 1

EXPERIMENT – 2

2. Write a Java program that implements Bubble sort algorithm for sorting in descending order and also shows the number of interchanges occurred for the given set of integers.

```
public class BubbleSort {
    static void sort(int[] a) {
        int interchange = 0;

        for (int i = 0; i < a.length - 1; i++) {
            for (int j = 0; j < a.length - 1 - i; j++) {
                if (a[j] < a[j + 1]) {
                    int temp = a[j];
                    a[j] = a[j + 1];
                    a[j + 1] = temp;
                    interchange++;
                }
            }
        }

        System.out.println("Number of interchanges: " + interchange);
    }

    public static void main(String[] args) {
        int[] a = {10, 5, 20, 8, 15};

        System.out.println("Array before sorting:");
        for (int i = 0; i < a.length; i++) {
            System.out.print(a[i] + " ");
        }
        System.out.println();

        sort(a);

        System.out.println("Array after sorting:");
        for (int i = 0; i < a.length; i++) {
            System.out.print(a[i] + " ");
        }
    }
}
```

OUTPUT:

```
Array before sorting:
10 5 20 8 15
Number of interchanges: 6
Array after sorting:
20 15 10 8 5
```

Figure 2: Output of program 2

EXPERIMENT – 3

3. Write a Java program that prompts the user for an integer and then prints out all the prime numbers up to that Integer.

```
import java.util.Scanner;

class PrimeNumbers {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      System.out.print("Enter a number: ");
      int num = sc.nextInt();

      System.out.println("Prime numbers from 1 to " + num + " are:");

      for (int i = 2; i <= num; i++) {
         boolean isPrime = true;

         for (int j = 2; j * j <= i; j++) {
            if (i % j == 0) {
               isPrime = false;
               break;
            }
         }

         if (isPrime) {
            System.out.print(i + " ");
         }
      }

      sc.close();
   }
}
```

OUTPUT:

```
Enter a number: 42
Prime numbers from 1 to 42 are:
2 3 5 7 11 13 17 19 23 29 31 37 41
```

Figure 3: Output of program 3

EXPERIMENT – 4

4. Write a Java program that checks whether a given string is a palindrome or not. Ex: MADAM is a palindrome.

```
import java.util.Scanner;

class Palindrome {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      System.out.print("Enter a string: ");
      String str = sc.next().toLowerCase();

      boolean isPalindrome = true;
      int j = str.length() - 1;
      for(int i = 0; i < str.length() / 2; i++) {
         if(str.charAt(i) != str.charAt(j--)) {
            isPalindrome = false;
            break;
         }
      }

      if(isPalindrome) {
         System.out.println(str + " is a palindrome.");
      } else {
         System.out.println(str + " is not a palindrome");
      }

      sc.close();
   }
}
```

OUTPUT:

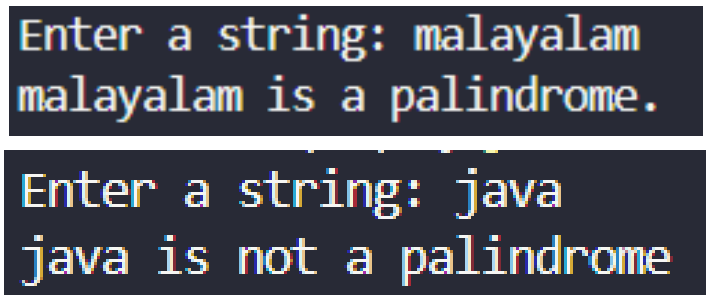


Figure 4: Output of program 4

EXPERIMENT – 5

5. Write a Java program that works as a simple calculator. Use a grid layout to arrange buttons for the digits and for the +, -, *, % operations. Add a text field to display the result. Handle any possible exceptions like divided by zero.

```
import java.awt.*;
import java.awt.event.*;
import javax.swing.JOptionPane;

public class Calculator implements ActionListener{
    Frame frame;
    TextField display;
    Button[] digitButtons = new Button[10];
    Button add, subtract, multiply, divide, clear, modulo, equalTo;
    double num1, num2, result;
    char operator;
    boolean calculationDone = false;

    public Calculator() {
        frame = new Frame("Calculator");
        frame.addWindowListener(new WindowAdapter() {
            @Override
            public void windowClosing(WindowEvent e) {
                frame.dispose();
                System.exit(0);
            }
        });

        display = new TextField();
        display.setEditable(false);
        display.setFont(new Font("Arial", Font.BOLD, 25));
        frame.add(display, BorderLayout.NORTH);

        Panel panel = new Panel();
        panel.setLayout(new GridLayout(4, 4));
        for (int i = 0; i <= 9; i++) {
            digitButtons[i] = new Button(String.valueOf(i));
            digitButtons[i].addActionListener(this);
        }

        add = new Button("+");
        subtract = new Button("-");
        multiply = new Button("*");
        divide = new Button("/");
        modulo = new Button("%");
        equalTo = new Button("=");
        clear = new Button("C");

        add.addActionListener(this);
```

```
        subtract.addActionListener(this);
        multiply.addActionListener(this);
        divide.addActionListener(this);
        modulo.addActionListener(this);
        equalTo.addActionListener(this);
        clear.addActionListener(this);

        panel.add(digitButtons[7]);
        panel.add(digitButtons[8]);
        panel.add(digitButtons[9]);
        panel.add(divide);

        panel.add(digitButtons[4]);
        panel.add(digitButtons[5]);
        panel.add(digitButtons[6]);
        panel.add(multiply);

        panel.add(digitButtons[1]);
        panel.add(digitButtons[2]);
        panel.add(digitButtons[3]);
        panel.add(subtract);

        panel.add(digitButtons[0]);
        panel.add(add);
        panel.add(modulo);
        panel.add(equalTo);

        frame.add(panel, BorderLayout.CENTER);

        frame.add(clear, BorderLayout.SOUTH);

        frame.setSize(400, 400);
        frame.setVisible(true);
    }

    @Override
    public void actionPerformed(ActionEvent e) {
        String command = e.getActionCommand();

        try {
            if(command.charAt(0) >= '0' && command.charAt(0) <= '9') {
                if (calculationDone) {
                    display.setText("");
                    calculationDone = false;
                    operator = '\0';
                }
                display.setText(display.getText() + command);
            }
```

```
        else if (command.equals("C")) {
            display.setText("");
            num1 = 0;
            num2 = 0;
            result = 0;
            operator = '\0';
            calculationDone = false;
        }

        else if (
            command.equals("/") ||
            command.equals("*") ||
            command.equals("+") ||
            command.equals("-") ||
            command.equals("%")
        ) {
            if (display.getText().isEmpty()) {
                    throw new NumberFormatException("Enter a number before choosing an
operator");
            }

            num1 = Double.parseDouble(display.getText());
            operator = command.charAt(0);
            calculationDone = false;
            display.setText("");
        }

        else if (command.equals("=")) {
            if (operator == '\0') {
                throw new NumberFormatException("Choose an operator");
            }

            if (!calculationDone && display.getText().isEmpty()) {
                throw new NumberFormatException("Enter some number first");
            }

            if (!calculationDone) {
                num2 = Double.parseDouble(display.getText());
            }

            switch(operator) {
                case '+':
                    result = num1 + num2;
                    break;
                case '-':
                    result = num1 - num2;
                    break;
                case '*':
                    result = num1 * num2;
```

```
                break;
            case '/':
                if(num2 == 0) {
                    throw new ArithmeticException("Cannot divide by zero");
                }
                result = num1 / num2;
                break;
            case '%':
                if(num2 == 0) {
                    throw new ArithmeticException("Cannot divide by zero");
                }
                result = num1 % num2;
                break;
        }

        display.setText(String.valueOf(result));
        num1 = result;
        calculationDone = true;
      }

    } catch (ArithmeticException ex) {
        JOptionPane.showMessageDialog(
            frame,
            ex.getMessage(),
            "Arithmetic Error",
            JOptionPane.ERROR_MESSAGE
        );
        display.setText("");
        return;

    } catch (NumberFormatException ex) {
        JOptionPane.showMessageDialog(
            frame,
            ex.getMessage(),
            "Input Error",
            JOptionPane.ERROR_MESSAGE
        );
        return;
    }
  }


  public static void main(String[] args) {
    new Calculator();
  }
}
```

OUTPUT:

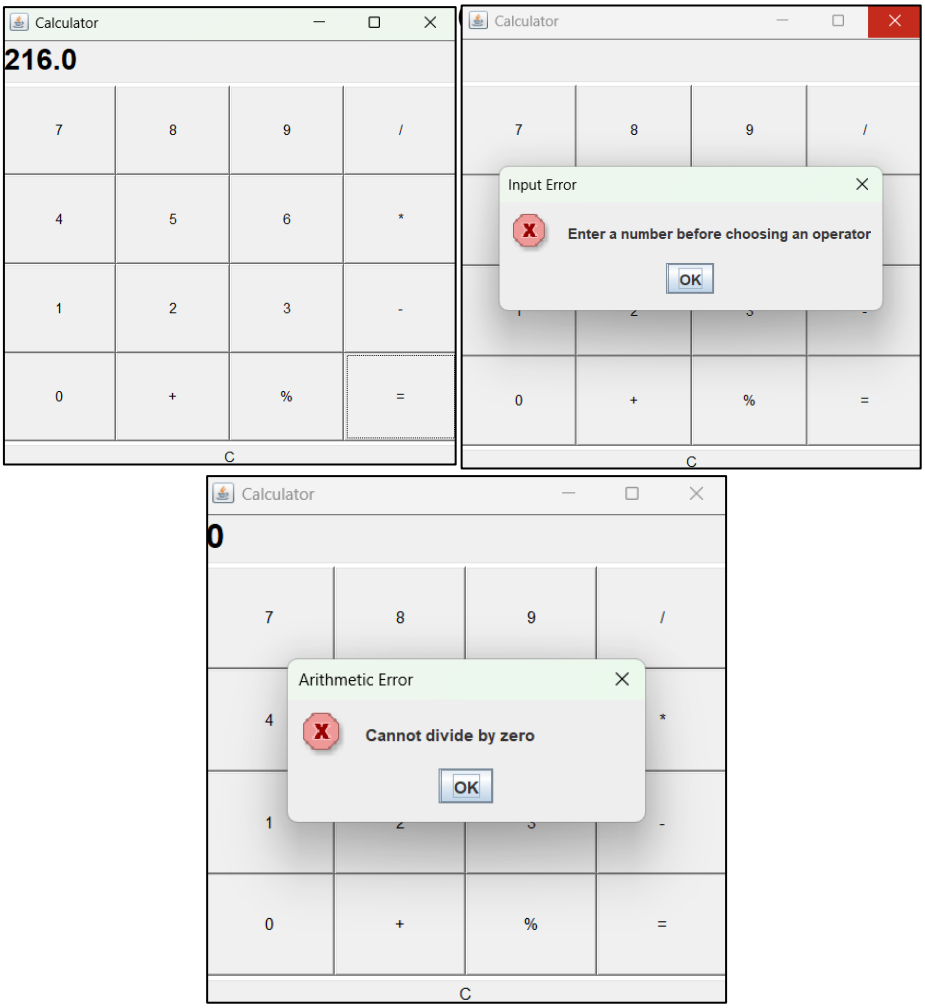


Figure 5: Output of program 5

EXPERIMENT – 6

6. Write a Java program that creates a user interface to perform integer divisions. The user enters two numbers in the text fields, Num1 and Num2. The division of Num1 and Num 2 is displayed in the Result field when the Divide button is clicked. If Num1 or Num2 were not an integer, the program would throw a Number Format Exception. If Num2 were Zero, the program would throw an Arithmetic Exception. Display the exception in a message dialog box.

```
import java.awt.*;
import java.awt.event.*;
import javax.swing.*;

public class IntegerDivision implements ActionListener {
   TextField num1TextField, num2TextField, resultTextField;
   Panel inputPanel, buttonPanel;
   Button integerDivision, equalsTo;
   int num1, num2, result;
   Frame frame;

   public IntegerDivision() {
      frame = new Frame("Integer Division");
      frame.addWindowListener(new WindowAdapter() {
         @Override
         public void windowClosing(WindowEvent e) {
            frame.dispose();
            System.exit(0);
         }
      });

      inputPanel = new Panel();
      inputPanel.setLayout(new GridLayout(1, 3));

      num1TextField = new TextField();
      num1TextField.setFont(new Font("Arial", Font.BOLD, 25));
      inputPanel.add(num1TextField);

      integerDivision = new Button("//");
      inputPanel.add(integerDivision);
      integerDivision.addActionListener(this);

      num2TextField = new TextField();
      num2TextField.setFont(new Font("Arial", Font.BOLD, 25));
      inputPanel.add(num2TextField);

      frame.add(inputPanel, BorderLayout.NORTH);

      buttonPanel = new Panel(new FlowLayout(FlowLayout.CENTER));
      equalsTo = new Button("=");
```

```
        equalsTo.setFont(new Font("Arial", Font.BOLD, 20));
        equalsTo.setPreferredSize(new Dimension(80, 40));
        buttonPanel.add(equalsTo);
        frame.add(buttonPanel, BorderLayout.CENTER);
        equalsTo.addActionListener(this);

        resultTextField = new TextField();
        resultTextField.setEditable(false);
        resultTextField.setFont(new Font("Arial", Font.BOLD, 25));
        frame.add(resultTextField, BorderLayout.SOUTH);

        frame.setSize(400, 170);
        frame.setVisible(true);
    }

    @Override
    public void actionPerformed(ActionEvent e) {
        String command = e.getActionCommand();

        if (command.equals("=")) {
            try {
                num1 = Integer.parseInt(num1TextField.getText().trim());
                num2 = Integer.parseInt(num2TextField.getText().trim());

                if (num2 == 0) {
                    throw new ArithmeticException(
                        "Cannot Divide by Zero."
                    );
                }

                result = num1 / num2;
                resultTextField.setText(String.valueOf(result));

            } catch (ArithmeticException ex) {
                JOptionPane.showMessageDialog(
                    frame,
                    ex.getMessage(),
                    "Arithmetic Error",
                    JOptionPane.ERROR_MESSAGE
                );
                resultTextField.setText("");

            } catch (NumberFormatException ex) {
                JOptionPane.showMessageDialog(
                    frame,
                    "Please enter valid integers.",
                    "Input Error",
                    JOptionPane.ERROR_MESSAGE
                );
```

```
                resultTextField.setText("");
            }
        }
    }

    public static void main(String[] args) {
        new IntegerDivision();
    }
}
```

OUTPUT:

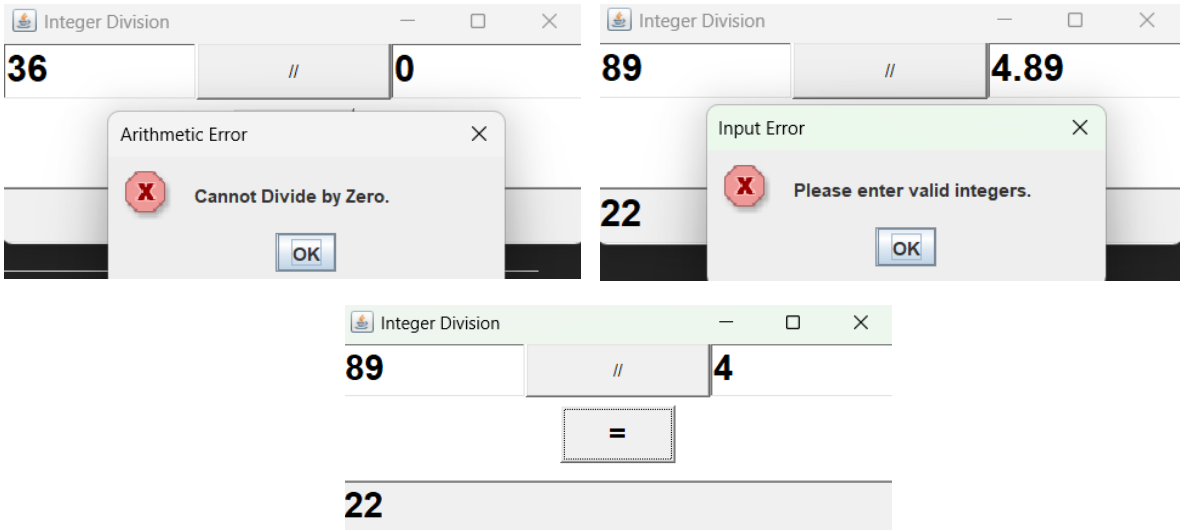


Figure 6: Output of program 6

EXPERIMENT – 7

7. Write a Java program for the following: Create a doubly linked list of elements. Delete a given element from the above list. Display the contents of the list after deletion.